

EFASOLAR Photovoltaic Inverters

Communication Guide



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The content of this document was checked and corresponds to the described hardware and software. However, since deviations cannot be precluded entirely, we cannot guarantee full consistency. The information given in this publication is reviewed at regular intervals and any corrections that might be necessary are made in the subsequent editions. Also, Efacec follows a continuous development policy and reserves the right to make changes and improvements in the product described in this document without previous notice.

1 Introduction

This document intends to be a guide for implementation of remote monitoring for the EFASOLAR Photovoltaic central Inverters.

These equipments are equipped with a communication interface that can be accessed using the MODBUS protocol.

There are available two options, MODBUS RTU or MODBUS TCP. For more information about the hardware and cable connection, please consult the equipment installation and operation manuals.

For more information about the MODBUS protocol please consult the website <http://www.modbus.org/>.

2 Configuration

The Photovoltaic Inverter is configured as a Slave node in the communication network and can be accessed by MODBUS RTU or MODBUS TCP. In both communication options the slave ID is set at 2 by default. For more information, please consult the inverter operation manual.

The Register Map is also described in the next sections and can be easily implemented on the MODBUS Master controller.

2.1 MODBUS RTU

In RTU (Remote Terminal Unit) Mode, each 8-bit message byte contains two 4-bit hexadecimal characters, and the message is transmitted in a continuous stream via a serial half duplex port. The settings for this interface are listed below:

Mode:	RTU
Baudrate:	9600 bps
Parity:	None
Data bits:	8 bits
Stop bits:	1 bits

Other settings can be configured by Efacec personnel, if requested.

2.2 MODBUS TCP

MODBUS TCP uses TCP/IP to carry the MODBUS messaging structure. On this mode, it should be defined the network settings (IP address, Default Gateway and Subnet Mask). To know more details about the network configuration, please consult the equipment manuals.

3 Register Map

3.1 Summary

The Inverter information is mapped in Holding Registers as follows:

Address	Var Type	Reg Idx	Units	Comments
41002 ...	float	Read	-	General measurements - Table 2
41051 41052 ...	uint16	Read	-	Alarms – Table 3
41056 ...	uint16	Read	-	Inverter Status – Table 4
41060 ...	float	Read	-	DC array current measurements - Table 5
41091 41094 ...	float	Read	-	Earth measurements – Table 6
41099 41832 ...	int32	Read	-	Optional energy analyzer – Table 7
41895 41702 ...	uint16	Read / Write	-	Inverter grid interface registers – Table 8
41713 41803	uint16	Read / Write	-	External commands – Table 9

Table 1 – Register Map

3.2 Inverter general measurements

Please note that the values labeled as float in the Var Type are encoded as single precision IEEE-754 big endian.

Address	Var Name	Var Type	Address	Reg Idx	Units	Comments
41002 41003	I _{DC}	float	high low	Read	A	DC Current
41004 41005	V _{DC}	float	high low	Read	V	DC Voltage
41006 41007	V _P	float	high low	Read	V	PV Voltage
41008 41009	P _{DC}	float	high low	Read	kW	DC Power
41010 41011	I _R	float	high low	Read	A	Phase R Current
41012 41013	I _S	float	high low	Read	A	Phase S Current
41014 41015	I _T	float	high low	Read	A	Phase T Current
41016 41017	V _{RS}	float	high low	Read	V	R-S Voltage ²
41018 41019	V _{ST}	float	high low	Read	V	S-T Voltage ²
41020 41021	V _{TR}	float	high low	Read	V	T-R Voltage ²
41022 41023	S	float	high low	Read	kVA	Apparent Power
41024 41025	P	float	high low	Read	kW	Active Power
41026 41027	Q	float	high low	Read	kvar	Reactive Power
41028 41029	Freq	float	high low	Read	Hz	Frequency
41030 41031	pf	float	high low	Read		Power Factor
41032 41033	ef	float	high low	Read	%	Efficiency
41034 41035	Energy_T	float	high low	Read	MWh	Total Energy Generated
41036 41037	Temp	float	high low	Read	°C	Inverter Temperature
41038 41039	Energy_D	float	high low	Read	kWh	Daily Energy Generated
41040 41041	Reserved	float	high low	Read		Reserved
41042 41043	Reserved	float	high low	Read		Reserved
41044 41045	Reserved	float	high low	Read		Reserved
41046 41047	Reserved	float	high low	Read		Reserved
41048 41049	Reserved	float	high low	Read		Reserved
41050 41051	Reserved	float	high low	Read		Reserved

²For EFASOLAR 100T these registers are phase-to-neutral voltage, instead of phase-to-phase voltage.

Table 2 – Holding Registers – Inverter Measurements

3.3 Inverter alarms

The next table shows all inverter alarms. (0 - is Inactive, 1- is Active).

Address	Mask	Comment
41052	0x0001	Any Alarm Active
41052	0x0002	High Panels Voltage
41052	0x0004	Low Panels Voltage
41052	0x0008	DC Current Peak
41052	0x0010	Low DC Current
41052	0x0020	Low Power
41052	0x0040	Low Pre-load Panels Voltage
41052	0x0080	High Grid Voltage
41052	0x0100	Low Grid Voltage
41052	0x0200	Grid Voltage Peak
41052	0x0400	Grid Current Peak
41052	0x0800	High Frequency
41052	0x1000	Low Frequency
41052	0x2000	Asymmetric Voltage
41052	0x4000	Asymmetric Current
41052	0x8000	Overload
41053	0x0001	Over Temperature Alarm
41053	0x0002	Internal Error 1
41053	0x0004	Stack Drivers
41053	0x0008	Internal Error 2
41053	0x0010	Surge Protection
41053	0x0020	Earth Leakage
41053	0x0040	Fan Failure
41053	0x0080	Input contactor error
41053	0x0100	Pre-load Contactor Error
41053	0x0200	Output contactor error
41053	0x0400	Over Temperature Error
41053	0x0800	Stack Driver R
41053	0x1000	Stack Driver S
41053	0x2000	Stack Driver T
41053	0x4000	Reserve
41053	0x8000	Low Pre-load Capacitors Voltage
41054	0x0001	High DC Bus Voltage
41054	0x0002	Low DC Bus Voltage
41054	0x0004	High Grid Current
41054	0x0008	Internal Error 3
41054	0x0010	Internal Error 4
41054	0x0020	Aux. Power Supply Fail
41054	0x0040	UPS Fail
41054	0x0080	DC Surge Protection
41054	0x0100	DC Fuses

41054	0x0200	Jammed
41054	0x0400	Locked
41054	0x0800	Internal Error 5
41054	0x1000	Reserve (Self Cons. comm.)
41054	0x2000	Reserve (Self Cons. analog power)
41054	0x4000	Reserve (Hybrid control. Comm.)
41054	0x8000	Array current monitor communication
41055	0x0001	Energy analyzer communication
41055	0x0002	GFDI fault
41055	0x0004	Reserve
41055	0x0008	Reserve
41055	0x0010	Reserve
41055	0x0020	Reserve
41055	0x0040	Reserve
41055	0x0080	Reserve
41055	0x0100	Reserve
41055	0x0200	Reserve
41055	0x0400	Reserve
41055	0x0800	Reserve
41055	0x1000	Reserve
41055	0x2000	Reserve
41055	0x4000	Reserve
41055	0x8000	Reserve

Table 3 – Holding Registers - Alarms

3.4 Inverter status

The Table 4 shows all inverter status registers available. Currently, only the register 41056 is used(0 – is Off, 1- is On).

Address	Mask	Comment
41056	0x0001	Inverter status
41056	0x0002	DC Contactor
41056	0x0004	AC Contactor
41056	0x0008	Grid Injection
41056	0x0010	-
41056	0x0020	-
41056	0x0040	-
41056	0x0080	-
41056	0x0100	-
41056	0x0200	-
41056	0x0400	-
41056	0x0800	-
41056	0x1000	-
41056	0x2000	-
41056	0x4000	-
41056	0x8000	-

Table 4 – Holding Registers – Inverter status

3.5 Inverter DC input array current measurements

The Table 5 shows the inverter additional Holding Registers used for DC array current measurements. As previously mentioned, these Holding Registers are only available in some EFASOLAR inverter models. For more information please consult the factory or the customer support.

Address	Var Name	Var Type	Address	Reg Idx	Units	Comments
41060 41061	lDC1	float	high low	Read	A	DC Current Array 1
41062 41063	lDC2	float	high low	Read	A	DC Current Array 2
41064 41065	lDC3	float	high low	Read	A	DC Current Array 3
41066 41067	lDC4	float	high low	Read	A	DC Current Array 4
41068 41069	lDC5	float	high low	Read	A	DC Current Array 5
41070 41071	lDC6	float	high low	Read	A	DC Current Array 6
41072 41073	lDC7	float	high low	Read	A	DC Current Array 7
41074 41075	lDC8	float	high low	Read	A	DC Current Array 8
41076 41077	lDC9	float	high low	Read	A	DC Current Array 9
41078 41079	lDC10	float	high low	Read	A	DC Current Array 10
41080 41081	lDC11	float	high low	Read	A	DC Current Array 11
41082 41083	lDC12	float	high low	Read	A	DC Current Array 12
41084 41085	lDC13	float	high low	Read	A	DC Current Array 13
41086 41087	lDC14	float	high low	Read	A	DC Current Array 14
41088 41089	lDC15	float	high low	Read	A	DC Current Array 15
41090 41091	lDC16	float	high low	Read	A	DC Current Array 16

Table 5 – Holding Registers – Array Current Measurements

3.6 Inverter earth measurements

Table 6 includes measurements related to the earth connection of the inverter.

Address	Var Name	Var Type	Address	Reg Idx	Units	Comments
41094	GND _{INS}	float	high	Read	kΩ	Earth insulation resistance measure
41095			low			
41098	GFDI _{LEAK} *	float	high	Read	A	Earth current leakage with pole to earth option (GFDI)*
41099			low			

*) GFDI current leakage measurement only if the inverter has the grounding kit option

Table 6 – Holding Registers – Grounding insulation and leakage measurements

As an Optional, the EFASOLAR Inverters may be equipped with the GFDI Automatic Control mechanism for the GFDI automatic measurement of the Earth Insulation (GND_{INS}). In this case, the following measurements are also available through Modbus:

Address	Var Name	Var Type	Address	Reg Idx	Units	Comments
41960	GND _{INS} [Last Measured Value]	float	high	Read	kΩ	Earth insulation resistance from last valid measure.
41961			low			
41164	GND _{INS} (Time since last measure)	float	high	Read	h	Time passed (in hours) since last measure was taken.
41165			low			

Table 7 – Holding Registers – Grounding insulation and leakage measurements

3.7 Optional energy analyzer

As an optional, an energy analyzer is integrated on the inverter for power/energy measurements at its output. The analyzer measurements are directly available through the inverter and described in table 7. The values are directly obtained from the analyzer and are available as 32-bit integers. Some of the measures are multiplied by a factor of 100 and must be divided to obtain the correct floating point value.

Address	Var Name	Var Type	Address	Reg Idx	Units	Comments
41832	ΣP	int32	high	Read	kW/100	Total real power +/-
41833			low			
41834	ΣQ	int32	high	Read	kvar/100	Total reactive power +/-
41835			low			
41836	ΣS	int32	high	Read	kVA/100	Total apparent power
41837			low			
41838	P1	int32	high	Read	kW/100	Real power phase 1 +/-
41839			low			
41840	P2	int32	high	Read	kW/100	Real power phase 2 +/-
41841			low			
41842	P3	int32	high	Read	kW/100	Real power phase 3 +/-
41843			low			

41844	Reserved			Read		
41845						
41846	Reserved			Read		
41847						
41848	Q1	int32	high	Read	kvar/100	Reactive power phase 1 +/-
41849			low			
41850	Q2	int32	high	Read	kvar/100	Reactive power phase 2 +/-
41851			low			
41852	Q3	int32	high	Read	kvar/100	Reactive power phase 3 +/-
41853			low			
41854	S1	int32	high	Read	kVA/100	Apparent power phase 1
41855			low			
41856	S2	int32	high	Read	kVA/100	Apparent power phase 2
41857			low			
41858	S3	int32	high	Read	kVA/100	Apparent power phase 3
41859			low			
41860	Reserved			Read		
41861						
41862	Reserved			Read		
41863						
41864	Hour	int32	high	Read	h/100	Operating Hours Counter h/100
41865			low			
41866	U12	int32	high	Read	V/100	Conductor voltage (1-2)
41867			low			
41868	U23	int32	high	Read	V/100	Conductor voltage (2-3)
41869			low			
41870	U31	int32	high	Read	V/100	Conductor voltage (3-1)
41871			low			
41872	V1*	int32	high	Read	V/100	Conductor voltage to N
41873			low			
41874	V2*	int32	high	Read	V/100	Conductor voltage to N
41875			low			
41876	V3*	int32	high	Read	V/100	Conductor voltage to N
41877			low			
41878	F	int32	high	Read	Hz/100	Frequency
41879			low			
41880	I1	int32	high	Read	A	Current phase 1
41881			low			
41882	I2	int32	high	Read	A	Current phase 2
41883			low			
41884	I3	int32	high	Read	A	Current phase 3
41885			low			
41886	In*	int32	high	Read	A	Current neutral
41887			low			
41888	ΣPF	int32	high	Read		Total power factor
41889			low			

41890	PF1	int32	high	Read		Power factor phase 1 -: Capacitive
41891			low			
41892	PF2	int32	high	Read		Power factor phase 2 -: Capacitive
41893			low			
41894	PF3	int32	high	Read		Power factor phase 3 -: Capacitive
41895			low			

*) Most inverter models have no AC neutral connection, so the values related to neutral have no real significance.

Table 8 – Holding Registers – Grounding insulation and leakage measurements

3.8 Inverter grid interface control registers

The Inverter Grid Interface Registers are listed in the table 8. These Holding Registers register are all unsigned integer 16 bits.

Address	Var Name	Var Type	Address	Reg Idx	Units	Comments
41702	QCtrlMode	uint16	-	Read/ Write	-	Reactive power control mode
41703	QCtrlPreference	uint16	-	Read / Write	-	Reactive or Active power preference
41704	QFPF_SP	uint16	-	Read / Write	-	Power factor setpoint
41705	QFQ_SP_%	uint16	-	Read / Write	%	Reactive power setpoint in %
41706	QFQ_SP_kVar	uint16	-	Read / Write	kvar	Reactive power setpoint in kvar
41707	QQuad_SP	uint16	-	Read / Write	-	Reactive power quadrant of operation setpoint
41708	PMaxCtrl	uint16	-	Read / Write	-	Maximum active power control mode
41709	PMax_SP_%	uint16	-	Read / Write	%	Maximum active power setpoint in %
41710	PMax_SP_kW	uint16	-	Read / Write	kW	Maximum Active power setpoint in kW
41711	PRedFreq	uint16	-	Read / Write	-	Instantaneous active power reduction with grid frequency
41712	PLimGrad	uint16	-	Read / Write	-	Limited gradient of active power increase after protective disconnection
41713	PLimGrad_%	uint16	-	Read / Write	-	Gradient of active power increase per minute

Table 9 – Holding Registers – Inverter Grid Interface Registers

For more information about Inverter Grid Interface Registers please consult the document: Mod. SESA 524 - EFASOLAR Grid Support Modbus Mapping.

It is advisable to consult Efacec to discuss the implementation of this interface and discuss the grid requirements.

3.9 Inverter external commands

The Inverter commands in the table 9 allows commanding the inverter remotely. The turn on / turn off maneuvers limit the expected lifetime of inverter components, care must be taken to limit the number of on/off operations of the inverter.

Address	Mask	Comment
41803	0x0001	Turn On
41803	0x0002	Turn Off
41803	0x0004	Reserved 1
41803	0x0008	-
41803	0x0010	-
41803	0x0020	-
41803	0x0040	-
41803	0x0080	-
41803	0x0100	-
41803	0x0200	-
41803	0x0400	-
41803	0x0800	-
41803	0x1000	-
41803	0x2000	-
41803	0x4000	-
41803	0x8000	-

Table 10 – Holding Registers – External Commands

To execute the instruction is necessary to set the bit to 1. After inverter acknowledgement this bit goes to 0. Confirm the status in the status registers after a few seconds.

4 Remote Monitoring System

The Register Map described on this document is a complete list of the information available on the Inverter Communication Interface.

It is advisable to consult Efacec to discuss the implementation of the Remote Monitoring System and selection of the most important data for this purpose.